

Conv_Simple — Live Anomaly Catalog

52 things MIRA can detect on the real bench, how to inject each one, and what Ask MIRA says

Audience: You, at the bench with the Micro 820 + GS10 rig | **Goal:** log live data, inject faults, prove each anomaly fires & Ask MIRA explains it

PLC: Allen-Bradley Micro 820 (2080-LC20-20QBB) @ 192.168.1.100:502 (Modbus TCP) | **PC:** 192.168.1.50 / Tailscale 100.72.2.99 | **Drive:** GS10 DURApulse, RTU slave 1, 9600/8N2

Signal source: plc/MbSrvConf_v4.xml (23 coils + 17 HRs) → plc/live-plc-bridge | **Engine:** plc/conv_simple_anomaly/ (A0-A12) | **Logger:** plc/conv_simple_anomaly/live_logger.py

Generated: 2026-06-12 | every row is grounded in a real signal — nothing invented.

What's in here

1. **The bench loop** — log → inject → confirm → ask, the five-step rhythm.
2. **The live signal inventory** — every coil & register we can read today.
3. **The 52-anomaly catalog** — by subsystem, with how to reproduce each at the bench.
4. **What Ask MIRA says** — the grounded answer shape per category.
5. **Two fixes this surfaces** — the reverse-cmd bug and the slave-map reflash.
6. **Suggested first capture sequence** + coverage summary.

1. The bench loop — how to run it through its paces

1. CCW: DISCONNECT from the controller, PLC in RUN (so the Modbus server answers)
2. Start a labeled capture for the fault you'll inject:
python plc/conv_simple_anomaly/live_logger.py --label flaky_photoeye
(type a note + Enter mid-run to stamp "broke beam now"; Ctrl-C to stop)
3. Inject the fault at the bench (see the "Reproduce" column below)
4. Stop the logger → logs/<stamp>_flaky_photoeye.{jsonl,csv} = your ground truth
5. Confirm the rule fires: python plc/conv_simple_anomaly/live_check.py --secs 6
6. Ask MIRA the question → it answers grounded in the same live tags

Capture baselines first

Record --label baseline_idle (everything stopped) and --label baseline_run_fwd before any fault. Every threshold — motor FLA, DC-bus band, frequency tolerance — is set *relative to those golden runs*. Without a baseline a number is just a number.

Read-only, always

The logger and live_check.py use only Modbus *read* function codes over Ethernet — side-effect-free per .claude/rules/fieldbus-readonly.md. Nothing here writes to the PLC or the GS10. Inject faults *physically* at the bench, never by writing registers.

2. Live signal inventory — what we can read today

Type	Signals (friendly names as logged)
Coils (23)	motor_running, conveyor_running, fault_alarm, vfd_comm_ok, system_ready, e_stop_active, dir_fwd, dir_rev, heartbeat, estop_wiring_fault, dir_fault, di00_fwd_sw, di01_rev_sw, di02_estop_nc, di03_estop_no, di04_pb_run, do00_green, do01_red, do02_contactor_q1, do03_pb_run_led, vfd_poll_active, vfd_fault_reset_pending, di05_photoeye ★
Registers (17)	motor_speed, motor_current, temperature, pressure, conveyor_speed, error_code, vfd_frequency_hz, vfd_current_a, vfd_voltage_v, vfd_dc_bus_v, item_count, uptime_seconds, conveyor_speed_cmd, conv_state, vfd_cmd_word, vfd_freq_setpoint, vfd_poll_step

★ `di05_photoeye` (coil 000023) is published only after the slave-map-v2 reflash — on the current live firmware it returns ILLEGAL DATA ADDRESS and the logger simply omits it.

Legend

Status: **LIVE** detectable now · **REFLASH** needs DI 5 / GS10-fault-reg reflash · **NEW** signal is live, rule not yet written · **A#** already coded in `rules.py` · **Sev:** **CRIT** **HIGH** **MED** **LOW** · **Alerts:** Telegram `/fault`, MCP `/api/faults/active`, ntfy push (HIGH/CRIT), Ask MIRA.

3. The 52-anomaly catalog

A. PLC & RS-485 / VFD comms health

#	ID	Anomaly	Signal(s)	Reproduce at bench	Sev	Status
1	A0	PLC / bridge offline	no fresh sample $\geq$ <code>offline_s</code>	Unplug PLC Ethernet / stop bridge	CRIT	A0 LIVE
2	A1	GS10 RS-485 link down	<code>vfd_comm_ok=0</code>	Unplug RS-485 / power off GS10	HIGH	A1 LIVE
3	B1	RS-485 flaky / intermittent	<code>vfd_comm_ok</code> toggles $\geq$ N	Partially seat / wiggle RS-485 plug	HIGH	NEW
4	B2	PLC heartbeat stalled (scan dead)	<code>heartbeat</code> not toggling $\geq$ X s	Put PLC in PROG / stop scan	CRIT	NEW
5	B3	VFD poll state-machine stuck	<code>vfd_poll_step</code> frozen / <code>vfd_poll_active</code> stuck=1	Half-fail comms mid-poll	MED	NEW
6	B4	PLC unexpected restart	<code>uptime_seconds</code> resets $\rightarrow$ 0	Power-cycle the PLC	HIGH	NEW
7	B5	Stale VFD values not trust-gated	<code>vfd_comm_ok=0</code> yet HRs look fresh	Pull RS-485, watch the HRs	MED	LIVE (gate)

B. Safety & E-stop (dual-channel)

#	ID	Anomaly	Signal(s)	Reproduce at bench	Sev	Status
8	A3	E-stop dual-channel wiring fault	<code>estop_wiring_fault=1</code> OR <code>di02_estop_nc==di03_estop_no</code>	Pull ONE e-stop channel wire	HIGH	A3 LIVE
9	S1	E-stop pressed (normal stop)	<code>e_stop_active=1</code>	Press the E-stop	INFO	LIVE
10	S2	E-stop NC channel stuck / jumpered	press, but <code>di02_estop_nc</code> stays 1	Jumper the NC channel	HIGH	LIVE
11	S3	E-stop NO channel stuck	<code>di03_estop_no</code> disagrees w/ NC	Jumper the NO channel	HIGH	LIVE
12	S4	E-stop chatter / flaky contact	<code>e_stop_active</code> toggles rapidly	Loosen e-stop wire, tap button	HIGH	NEW
13	A5	Belt running while not permitted	<code>motor_running</code> $\wedge$ (<code>estop</code> $\vee$ <code>wiring</code> $\vee$ <code>¬contactor_q1</code>)	Force run with a safety open	HIGH	A5 LIVE
14	S5	Contactor Q1 dropped under run cmd	<code>do02_contactor_q1=0</code> $\wedge$ run	Open contactor coil while running	HIGH	NEW

C. Direction & control logic

#	ID	Anomaly	Signal(s)	Reproduce at bench	Sev	Status
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15	A4	Direction conflict (both selected)	<code>di00_fwd_sw ∧ di01_rev_sw (dir_fault)</code>	Throw BOTH direction switches	MED	A4 LIVE
16	C1	Reverse-run blind spot (bug)	<code>vfd_cmd_word=34 not in run_cmd_values [18,20]</code>	Run conveyor in REVERSE	FIX	config → [18,34]
17	A6	Drive not responding to RUN	<code>cmd_word∈{18,34} ∧ ¬motor_running ≥ grace</code>	Command run, GS10 disabled	MED	A6 LIVE*
18	C2	Run command with no direction	<code>run word ∧ ¬ dir_fwd ∧ ¬ dir_rev</code>	Command run, no dir switch	MED	NEW
19	C3	Direction reversed under load	<code>dir_fwd→dir_rev while motor_running</code>	Flip dir switch while running	MED	NEW
20	C4	Run pushbutton stuck / held	<code>di04_pb_run held > X s</code>	Tape the run button down	LOW	NEW

D. Drive electrical (GS10 — current / voltage / DC-bus / frequency)

#	ID	Anomaly	Signal(s)	Reproduce at bench	Sev	Status
21	A8	Output overcurrent > FLA	<code>vfd_current_a > motor fla_a</code>	Brake / load the belt	HIGH	A8 LIVE
22	D1	Locked-rotor / stall	<code>high vfd_current_a ∧ frequency_hz>0 ∧ motor_speed=0</code>	Hold the belt still	HIGH	NEW
23	A9	DC bus undervoltage	<code>vfd_dc_bus_v < dc_bus_lo_v</code>	Input sag (hard to force)	MED	A9 LIVE
24	A9	DC bus overvoltage (regen)	<code>vfd_dc_bus_v > dc_bus_hi_v</code>	Rapid decel of loaded belt	MED	A9 LIVE
25	A10	Output frequency frozen under RUN	<code>frequency_hz unchanged ≥ freq_frozen_s</code>	Drive stops ramping	MED	A10 LIVE
26	A7	Output Hz not tracking setpoint	<code> frequency_hz - vfd_freq_setpoint > tol</code>	Change setpoint, drive lags	MED	A7 LIVE
27	A2	GS10 drive fault active (decoded)	<code>fault_raw (0x2100) → GS10_FAULT_CODES</code>	Trip GS10 (oC / oL / oH / E.LV)	HIGH	REFLASH
28	D2	Output current zero under run	<code>vfd_current_a≈0 ∧ motor_running ∧ Hz>0</code>	Open a motor lead / blow output fuse	HIGH	NEW
29	D3	Output voltage zero with run cmd	<code>vfd_voltage_v≈0 ∧ run</code>	Output-stage fault	HIGH	NEW
30	D4	Undercurrent / load lost	<code>vfd_current_a << baseline while running</code>	Decouple belt from motor	MED	NEW
31	D5	Current spike transient	<code>Δvfd_current_a > jump in one interval</code>	Brief jam pulse	MED	NEW

E. Photo-eye & material flow (most need the DI 5 reflash)

#	ID	Anomaly	Signal(s)	Reproduce at bench	Sev	Status
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32	A12	Photo-eye jam (continuous block)	<code>di05_photoeye</code> blocked $\geq X$ s	Hold the beam broken	HIGH	REFLASH
33	E1	Photo-eye flaky / rapid toggle (#1668)	<code>di05</code> ≥ 7 falling edges in ~ 5.6 s	Wiggle sensor / wave hand fast	HIGH	REFLASH
34	E2	Photo-eye dead / dark (never makes)	<code>di05</code> never trips over a run	Disconnect the sensor	MED	REFLASH
35	E3	Photo-eye stuck-made (never breaks)	<code>di05</code> never clears	Block / misalign permanently	MED	REFLASH
36	E4	Photo-eye loose wire (intermittent open)	<code>di05</code> brief random drops	Loosen the terminal	MED	REFLASH
37	E5	Unexpected trip while stopped	<code>di05</code> changes $\wedge$ <code>conveyor_running=0</code>	Wave hand at eye while off	LOW	REFLASH
38	E6	No item flow while running	<code>item_count</code> flat $\wedge$ running $\wedge$ motion	Run an empty belt	MED	NEW
39	E7	Item jam (flow + beam)	<code>di05</code> blocked $\wedge$ <code>item_count</code> frozen	Jam an item at the eye	HIGH	REFLASH

F. Motion & speed

#	ID	Anomaly	Signal(s)	Reproduce at bench	Sev	Status
40	F1	Commanded run, no motion	<code>run cmd</code> $\wedge$ <code>conveyor_speed</code> ≈ 0 past grace	Belt slips / drive disabled	HIGH	NEW
41	F2	Motion while commanded stop	<code>conveyor_speed</code> > 0 $\wedge$ stop cmd	Excess coast after stop	MED	NEW
42	F3	Speed not reaching command	$ \text{conveyor_speed} - \text{speed_cmd} > \text{tol}$ steady	Overload the belt	MED	NEW
43	F4	Belt overspeed	<code>conveyor_speed</code> $> \text{speed_cmd} + \text{tol}$	(drive misconfig)	MED	NEW
44	F5	Speed hunting / oscillation	high variance in <code>conveyor_speed</code>	PID instability	LOW	NEW

G. State machine, process & HMI consistency

#	ID	Anomaly	Signal(s)	Reproduce at bench	Sev	Status
45	G1	Fault alarm latched	<code>fault_alarm=1</code>	Trigger any ladder fault	HIGH	NEW
46	G2	Error code nonzero (decode)	<code>error_code</code> $\neq 0$	Trip a fault path	var	NEW
47	G3	<code>conv_state</code> wedged in FAULT	stuck non-running $\geq X$	Trip, then don't reset	MED	NEW
48	G4	System not ready w/o cause	<code>system_ready=0</code> $\wedge$ no estop/fault cause	Remove a start precondition	MED	NEW
49	G5	Inconsistent state	<code>conveyor_running</code> $\wedge$ <code>¬system_ready</code> (or <code>motor</code> $\wedge$	Force contradictory states	MED	NEW

			~conveyor)			
50	G6	Stack-light logic mismatch	do00_green $\wedge$ do01_red both / neither	Diagnostic check	LOW	NEW
51	G7	Motor over-temperature	temperature > limit	Heat sensor / sustained load	HIGH	NEW
52	G8	Temperature rising trend (predictive)	temperature slope > X over window	Gradual loading	MED	NEW

4. What Ask MIRA says

Each answer leads with **equipment + fault state**, then **likely cause**, then **next check** — never a raw-tag wall (per the industrial-HMI rule). Representative answers:

Category	Ask MIRA reply (grounded shape)
A1 / B1 comms	"GS10 isn't answering on RS-485 — its values are stale, treat them with suspicion. Likely a loose / re-seated RS-485 conductor or termination. Check the A/B pair at both ends; confirm GS10 is 9600 8N2, slave 1."
A3 / S2 e-stop	"E-stop channels disagree (NC vs NO) — one contact or wire has failed. This can defeat the stop. Verify both channels switch together; check the suspect leg."
A8 / D1 current	"Output current is above motor FLA with little/no motion — mechanical jam, seized bearing, or overloaded belt. Stop and clear the obstruction before restarting."
A12 / E1 photo-eye	"PE-101 is chattering (7 fast breaks in ~6 s) while the line is clear — a flaky sensor: loose connector, marginal alignment, or a wiring fault. Reseat and realign."
G1 / G2 fault latch	Decodes <code>error_code</code> / GS10 fault code → plain-language cause + reset steps.

5. Two fixes this catalog surfaces

1 — Reverse cmd-word bug (C1)

`config.yaml` has `run_cmd_values`: `[18, 20]` but REV+RUN on the GS10 is **34** (0x20 REV + 0x02 RUN), which you verified at the bench. Until fixed, rules A6 (drive-not-responding) and A10 (freq-frozen) are *blind whenever the belt runs in reverse*. One-line change to `[18, 34]` plus a rule test.

2 — Slave-map-v2 reflash

Unblocks every **REFLASH** row at once — the whole photo-eye family (E1–E7) plus A2 GS10-drive-fault decode. The committed map adds DI 5 (coil 000023); the live PLC hasn't been reflashed yet (`coil 23 → ILLEGAL DATA ADDRESS`). See `plc/RESUME_1668_PLC_FEED.md` §3.

6. Suggested first capture sequence + coverage

Run these, in order — all live, no reflash needed

1. `--label baseline_idle` — 30 s, everything stopped
2. `--label baseline_run_fwd` — run forward
3. `--label baseline_run_rev` — run reverse (confirms `vfd_cmd_word=34` in the data)
4. `--label estop_press` → `--label estop_wiring_fault` (pull one channel)

5. `--label rs485_flaky` — wiggle the serial plug
6. `--label belt_stall` — hold the belt, watch `vfd_current_a` climb

Coverage	Count	Notes
LIVE today (no reflash)	~30	All of A / B / C / D-electrical / F / G groups
Needs DI 5 / fault-reg reflash	8	Photo-eye family E1–E7 + A2 drive-fault decode
Already coded (<code>rules.py</code>)	12	A0–A12
NEW candidates to write	~30	B1–B4, S4/S5, C2–C4, D1–D5, E6, F1–F5, G1–G8

Source of truth: `plc/conv_simple_anomaly/ANOMALY_CATALOG.md` · Logger:
`plc/conv_simple_anomaly/live_logger.py` · Engine: `plc/conv_simple_anomaly/rules.py` · Slave map:
`plc/MbSrvConf_v4.xml` . Every signal above is read-only Modbus TCP — nothing here writes to the PLC.